ESE 4013 One-Shot Notes

Energy in Built Environment · KUET · past papers 2020, 2021, 2022, 2024 · answers from the course slides

Section A: Built environment, climate, sun, wind, shading, daylight

Building and energy

Envelope = 5 elements: foundation, walls, roof, windows, doors (separates in/out). **Fabric** = the materials/components it is made of.

Buildings: ~40 % of world energy, ~39 % of CO_2; people indoors 80-90 % of life.

Life-stage energy: **embodied, operational, maintenance, demolition**. End use: HVAC 35 %, lighting 11 %, appliances 18 %, other 36 %.

Six factors: **climate, envelope, systems, O&M, IEQ, occupants** (+ social).

IEQ and IAQ

IEQ = **IAQ + thermal + visual + acoustic**. Poor IEQ: eye/nose/throat irritation, eczema, headache, fatigue, low concentration, sick leave.

Contaminant sources: **outside, equipment (HVAC/non-HVAC), human activities, building components/furnishings, other**. Control: source → ventilate (7.5 L/s/person, CO_2 <1000 ppm) → filter → measure.

Comfort

Thermal comfort (ASHRAE): state of mind expressing satisfaction with the thermal environment. 4 environmental (T, MRT, air speed, RH) + 2 personal (met = 58.2 W/m², clo = 0.155 m²K/W).

Visual comfort: subjective visual well-being; task lighting, local control, no over-illumination, daylight.

Noise: sound that disturbs silence/intended sound, annoys or harms. Acoustic comfort by **insulation** (mass, airtight) + **absorption** (porous finishes); reverberation time, dB(A), STC, NRC.

Saving energy (7 areas)

Heating, cooling, ventilation, lighting, hot water, equipment, good housekeeping. Lighting: daylight, efficient luminaires, controls, sun-pipes; personal control saves 10-20 %.

New building: orient, form, envelope, window, shade, daylight, ventilate, efficient systems, renewables.

Sun-earth

Rotation 24 h → day/night. **Revolution** 365¼ d + 23½° tilt → seasons.

Solstice: axis farthest toward/away (21 Jun over Tropic of Cancer; 21 Dec over Capricorn). **Equinox**: sun over equator, day = night (21 Mar, 23 Sep).

Weather = short-term local; climate = long-term average.

```
1367 W/m2 outside atmosphere: 26 %
reflected, 18 % absorbed, 56 % reaches
ground
G = B + D + R   (diffuse 10-20 % clear,
up to 100 % overcast)
```

Bangladesh: long axis **E-W**, glazing N/S, shade south, avoid west glass.

Sky temperature

```
Tsky = Ta - 6 ;  Swinbank Tsky = 0.0552
Ta^1.5 (K)
Lclear = 5.31e-13 Ta^6 ; Lcloud =
ec*sigma*(Ta-5)^4
Lsky = (1-c)Lclear + c*Lcloud ; c = (Kd-
0.165)/0.835 ; Tsky = (Lsky/sigma)^0.25
```

Clear night: big radiative loss (roof colder than air); cloudy: small.

Urban effects

UHI: city hotter than country: waste heat, dark low-albedo surfaces, less green/water, canyons trap radiation, low wind.

Urban wind island: city wind obstructed, deflected, sometimes faster (around sunrise/sunset); roughness, down-wash, Venturi, canyon vortex.

Wind patterns (increase)

Wind shadow 3-4× building size. Bar 1.4×; Venturi 1.3× (30 m), 1.6× (50 m); gap 1.2× (7 storeys), 1.5× (60); corner 1.2× (15 m), 1.5× (>35 m), 2.2× between 100 m towers; wake 1.4-2.2×; gap in windbreak 100-120 %.

Ventilation: face the wind, inlet low/outlet high, outlet ≥ inlet, wing walls, thin plan, avoid wind shadows.

Shading

Internal (glare), external (best, up to 80 % cut), interpane. **South** = horizontal; **E/W** = vertical/movable; hot climate = egg-crate; north little; deciduous trees.

```
HSA = wall azimuth - solar azimuth
(fins)
tan VSA = tan(altitude)/cos(HSA)
(overhang)
Overhang depth D = H/tan VSA ; side
extension = D tan HSA
```

Critical: Sept noon (S), Sept 10 am (E), Sept 3 pm (W). Fulbarigate Sept noon: alt 67°, az −8° → VSA 67.2°, D = 0.9/2.379 = 0.38 m, width 0.61 m.

Daylighting

Controlled admission of sunlight + skylight to cut electric light. **Passive** (fixed) vs **active** (tracking mirrors, GPS, light well, diffuser). Top: skylight, monitor, saw-tooth (faces north), clerestory, tubular. Side: windows, light shelf.

Roof light = 3× window per m², ≤ 12 % of floor. Glazing ratio 25-50 %. Savings 15-40 %. Higher window head → deeper light. Office sill 1-1.2 m; house 0.7-0.9 m.

Glare: excessive brightness contrast; discomfort and disability.

Six principles: orientation, form, glazing ratio, window height, overhead, redirection.

Section B: Thermal performance, loads, annual energy, NZEB

Windows

```
Uw = (Ucg Acg + Ueg Aeg + Uf Af)/Aw
Single: R = 1/hi + L/k + 1/ho =
0.121+0.003+0.029 = 0.153 (glass 2 %)
Double: 1/U = 1/hi + 1/hspace + 1/ho ;
e_eff = 1/(1/e1 + 1/e2 - 1)
```

Air k = 0.025 (1/30 of glass). Gap heat: half radiation (low-e, ε 0.1 halves it), half conduction (optimum ~13 mm, then convection). Argon/krypton, triple.

Cold: high solar τ, low-e for far IR, low U. Hot: visible in, IR out (absorbing film outside, low-e inside), −60 % solar gain, −15-30 % cooling load.

Selective low-e also reflects near-IR (low SHGC + high VT).

SHGC

```
tau + rho + alpha = 1
SHGC = solar gain / incident solar
Q/A = U(Ti-To) + SHGC*IT
single: SHGC = tau + alpha*U/ho ,  U =
hi*ho/(hi+ho)
double: SHGC = tau + ao*U/ho + ai*U(1/ho
+ 1/hs)
```

Low SHGC not always best: cold climates and south sun want gain; daylight; orientation.

Smart windows

Thermochromic (temperature), photochromic (light), liquid crystal (voltage, privacy), electrochromic (voltage, controllable). Bangladesh: **thermochromic** (follows heat, not light).

Sol-air

```
Tsa = Tamb + as*q/ho - e*sigma*(Tamb^4 -
Tsurr^4)/ho
Q = U A (Tsa - Ti) ; dTsolar = as*q/ho
solar fraction = (Tsa - Tamb)/(Tsa - Ti)
fraction of incident solar inside =
U*as/ho
as/ho: light 0.026, dark 0.052 m2K/W (ho
= 17)
```

Radiation term ~0 walls, ~4 °C roofs. 2021: 1.8×20×26 = **936 W**; fraction 13/26 = **50 %**.

Internal gains

```
Lights: Q = PL * FUL * FSA   (FSA 1.2
fluorescent)  or  LPD * area
Equipment: Q = (PM/etaM) * FUM * FLM
Infiltration: Qs = rho*cp*V*(Ti-To) ; Ql
= rho*hfg*V*(Wi-Wo), rho*hfg = 3010
W/(L/s)
```

Person 100 W rest to 500 W heavy; latent ≥ 30 W (1 L water/day). Latent = instant; sensible delayed by mass.

Transient

```
dT/dt = alpha d2T/dx2 , alpha = k/(rho
cp)
Interior: Tm' = Fo(Tm-1 + Tm+1) + (1-
2Fo)Tm , Fo = alpha dt/dx^2 ⩽ 1/2
Surface: T0' = 2Fo(T1 + Bi*Tinf) + (1 -
2Fo - 2Bi*Fo)T0 , Bi = h dx/k
h = hc + hr , hr = e*sigma*(T0+Ts)
(T0^2+Ts^2)
CTF: Qt = -sum dn Q(t-n) + A[sum bn
Tos(t-n) - Trc sum cn]
```

4 reasons: envelope mass, stored solar, internal schedules, controls. Heavy wall: lag 6-12 h, smaller peak.

Infiltration

```
ACH = V(m3/h) / Volume ; min 0.35 ACH,
7.5 L/s per person
Crack: V = C dp^n (n ~0.65) ; orifice V =
CD A sqrt(2dp/rho)
dp = dp_wind + dp_stack + dp_vent ;
dp_wind = Cp rho v^2/2
dp_stack = -rho_i g dh (Ti-To)/To  (x Cd)
LBL: V = Aleak sqrt(as dT + aw v^2) ;
tracer: N = ln(C0/C)/t
```

Stack (winter): in at bottom, out at top, neutral plane mid-height. Prevent: caulk fixed joints, weather-strip moving joints.

Annual energy

```
Tbal = Ti - Qgain/Ktot ; Ktot = sum UA +
rho cp V = Qdesign/(Ti-To)des
HDD = 1 day * sum(Tbal - To)+ ; Qyr =
Ktot * HDD * 24 / eta
CDD = sum(To - Tbal)+ ; HDD - CDD = 365
(Tbal - To_avg)
Tmax = Ti - Qgain/Kmax ; Qc =
Ktot[CDD(Tmax) + (Tmax-Tbal)Nmax]/eta
Bin: Q = sum Nbin * Ktot/eta * (Tbal -
To)+
```

Peak load = worst weather, sizes equipment; annual = average weather + efficiency.

Ground and zoning

```
Tsoil(0,t) = Tavg + dT sin(2 pi t/8760 -
phi - pi/2)
Basement wall: Q = Uavg,bw * Pb * hbw *
(Ti - Tsoil)  (radial paths)
Slab: Q = Fp * P * (Ti - To)
Tunheated = [(UA)c Ti + (UA)u To]/[(UA)c
+ (UA)u]
```

Zone = one thermostat. Long narrow plan > square in heat loss (more perimeter).

NZEB and green

US DOE: produces its own annual energy by renewables. EU: **nearly** ZEB, very high performance, very low need mostly renewable on/near site. Types: **site, source, cost, emission**.

Design order: **envelope → efficiency → renewables** (+ commissioning 5-10 %). LEED: register, prerequisites + credits, submit, GBCI review; Certified 40, Silver 50, Gold 60, Platinum 80.

Desiccant and evaporative

Direct evaporative: cooler + more humid (dry climates). Indirect: cools at constant W. Desiccant (silica gel, alumina, LiCl, TEG), regenerate 60-90 °C by evacuated-tube solar. Sequence: desiccant wheel → thermal wheel → evaporative cooler; exhaust: evap → thermal wheel → solar heater → desiccant → out. Hot-humid NZEB: desiccant DOAS for latent + high-temperature radiant/chilled beam for sensible + PV.

Night cooling offsets 20-30 W/m² (natural), 50 W/m² (mechanical), night < 20 °C.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ACH	air changes per hour
ALT	solar altitude angle
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BEMS	building energy management system
CDD	cooling degree-days
COP	coefficient of performance
CTF	conduction transfer function
dB	decibel (sound level unit)
DF	daylight factor
EUI	energy use intensity
HDD	heating degree-days
HSA	horizontal shadow angle
HVAC	heating, ventilation and air-conditioning
IAQ	indoor air quality
IEQ	indoor environmental quality
LC	liquid crystal
LED	light-emitting diode
LEED	Leadership in Energy and Environmental Design (green building rating)
NRC	noise reduction coefficient
NZEB	net-zero energy building
PMV	predicted mean vote (comfort index)
PPD	predicted percentage dissatisfied
PV	photovoltaic (solar cells)
SC	shading coefficient
SF	solar factor
SHGC	solar heat gain coefficient
STC	sound transmission class
UHI	urban heat island
VOC	volatile organic compound
VOCs	volatile organic compounds
VSA	vertical shadow angle
VT	visible transmittance